

Alexander Marshall | Physicist

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Royster Fellow | Doctoral Candidate | UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics <i>The University of North Carolina at Chapel Hill</i> Royster Doctoral Fellow Exams Passed All But Dissertation	Chapel Hill, NC 2027
Master of Science in Physics <i>The University of North Carolina at Chapel Hill</i> Completed <i>en route</i> to PhD	Chapel Hill, NC 2025
Physics B.S. Philosophy B.A. Mathematics Minor <i>Wake Forest University</i> GPA 3.91 <i>Summa Cum Laude</i> Honors in Physics Stamps Scholar	Winston-Salem, NC 2022

Academic Projects

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSMF with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Undergraduate Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands

of phase settings on a spatial light modulator. Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Select Additional Research.....

Chromatin Structural Evolution Alters Histone Transport Dynamics Chapel Hill, NC
UNC-Chapel Hill, Dr. Kerry Bloom 2025

Tracking and analyzing the trajectories of HALO-tagged histones during metaphase and interphase to probe structural changes occurring in chromatin.

Enabling Low-Resource Microscopy Chapel Hill, NC
UNC-Chapel Hill, Dr. Richard Superfine 2023-Present

Supervised a primarily undergraduate-driven effort to develop cutting edge microscopy techniques for use in low-research environments. Undergraduates won prizes at annual UNC MakerFest competition for two consecutive years.

Employing Advanced LSM for Enhanced Imaging Chapel Hill, NC
UNC-Chapel Hill, Dr. Richard Superfine 2022-Present

Engineered an advanced optical system to conduct new methods of LSM, such as HILO and HIST. Due to software rot, had to rebuild system from the ground up, involving extensive design and engineering. Also implemented new optical modules and DAQ-board mediated software to enable sub-millisecond automated control of optical hardware.

Neurochemical Addiction Mechanisms in the Nucleus Accumbens Winston-Salem, NC
Wake Forest School of Medicine, Dr. Evgeny Budygin 2019-2021

Conducted cyclic voltammetry on rats *in vivo* and mouse brain slices *in situ* to explore the role of neurotransmitters in addictive behaviors.

Cultural Impact of Oenology in the Peloponnese Winston-Salem, NC
Wake Forest Department of Anthropology, Dr. Karen Friederic 2019

Conducted a month-long anthropological study in Nemea, Greece on how the economic significance of the viticultural industry impacts local culture. I found that extreme seasonal workloads intersect with traditional gendered domestic labor expectations to inhibit female career growth in this industry.

Scholarships and Grants

Royster Doctoral Fellowship Chapel Hill, NC
Royster Foundation, UNC-Chapel Hill Graduate School 2022-2027

Five-year doctoral recruitment fellowship. Provides extradepartmental funding for the majority of a five year doctoral program.

Stamps Scholarship Winston-Salem, NC
Stamps Foundation 2018-2022

Four-year scholarship awarded for academic merit. Covered all costs of attendance to Wake Forest University, including room and board. Also provided a technology grant and semesterly stipend.

Talks and Symposia

Physiological Strains in Poroelastic Media Induce Asymmetric Transport Chapel Hill, NC
Master's Defense, UNC-Chapel Hill November 17, 2025

Strain Alters Anomalous Diffusion in Poroelastic Media. Elon, NC
Symposium Talk, Elon University October 1, 2025

Diffusion in Strained Poroelastic Media Chapel Hill, NC
Thesis Proposal, UNC-Chapel Hill September 16, 2025

Generating a 3-D Lattice of Photonic Spheres Winston-Salem, NC
Honors Thesis Defense, Wake Forest University May 2022